

J4/J5 control and evidence correction

TASK-20261007-efaa15 producer

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Corrected scope and mathematical boundary

[Derived; producer submission] This new packet corrects the control semantics and source/visual evidence rejected in BATCH-9681e3. It supplies no fresh J4/J5 verdict. J1-J3 remain the reviewed historical conditional statements, not new findings.

The fixed symbolic slice is $p > 3$ prime, $p = 3 \pmod{4}$, $q = p^2$, $E : y^2 = x^3 + x$, $D = 2^a$, $N = 5^b > 4D$, $a \geq 2$, $b \geq 1$, $v_2(p+1) = a$, $v_5(p+1) = b$, and $k = \lfloor a/2 \rfloor$. Public keys bind ordered roles (E, P, U, V) ; ciphertexts are exact short-Weierstrass model C plus full signed X . No concrete sampled endpoints exist.

For $s = m + 2^k r \in [0, D)$, the honest relation uses the normalized separable degree- D map $f : E \rightarrow C$ with kernel $\langle U + sV \rangle$ and $f(P) = X$. Normalization means $f^* \omega_C = \omega_E$, with $\omega = dx/(2y)$ on each fixed model. The sender's fixed degree-two formulas determine exact coefficients at every step.

[Retrieved + derived] Hasse's primary norm-addition formula, with degree multiplicativity, transfers from endomorphisms to $\text{Hom}(E, C)$: compose f, g with one fixed nonzero $u : C \rightarrow E$ and cancel $\deg u$. Hence

$$\deg(f + g) + \deg(f - g) = 2 \deg f + 2 \deg g, \quad \deg(f \pm g) \leq 4D.$$

A nonzero map killing a point of order M has $M \leq \deg_s(h) \leq \deg(h)$. Its difference need not be separable. These bounds compare maps on the same exact source and target.

[Correction] Equal x-only observations give $g(P) = \pm f(P)$. The bound then gives $g = f$ or $g = -f$. The latter violates simultaneous normalization in $p > 3$, since $[-1]^* \omega_C = -\omega_C$. Even without normalization, f and $-f$ have the same kernel. Thus the old normalized cross-message collision claim is withdrawn. Full signed X remains the original serialization rule; this example does not establish its necessity for message injectivity under these hypotheses.

[Correction] $P = O$ or lower-order P removes the $N > 4D$ proof premise. The applicable bound uses $M = \text{ord}(P)$: uniqueness still holds for $M > 4D$. Failure to derive it when $M \leq 4D$ does not exhibit a collision. Both P and X exact-order tests must be weakened for the corresponding honest objects to pass.

[Limit] There are no measured data or runs. Source checking, symbolic derivation, parsing and rendering are documentation checks. BaseGen, exact recipient map and uniform Eval2 remain open.

Controls with the exact implication

The standalone blind statement contains the submitted definitions, parameters, classifications and control claims without derivations. The control matrix adds complete acceptance traces. All examples fix one key before comparing messages.

C1/C2 - proof-premise controls. Zero or lower-order P passes weakened membership, and the same weakening for X is required. Full admission rejects the penultimate-multiple order tests. No same-exact-C different-message pair is claimed. For $b = 1$ there is no nonzero lower-order 5-power point.

C3 - exact symbolic witness W_V_ZERO. Take one key (E, P, U, O) with $\text{ord}(U) = D$, $\text{ord}(P) = N$. Let f_U be the fixed normalized quotient by $\langle U \rangle$, C its exact coefficient output, and $X = f_U(P)$. Messages 0,1 with $r=0$ have identical kernels and use the identical map and exact ciphertext. Membership-only V and pairing tests accept; exact V order and primitive pairing reject. All retained P/X and U exact orders, model/count checks, degree D and normalization hold conditional on the original valid BaseGen slice.

C4/C5 - exact symbolic witness W_DEPENDENT. Take one key (E, P, U, U) , $a \geq 4$, $k \geq 2$. Messages 0,2 with $r=0$ give U and $3U$, generating the same exact subgroup. At every quotient stage its unique order-two subgroup is unchanged, so the identical normalized formulas give the same exact C and $f_U(P)$. Both U,V have exact order D, but $e_D(U, U) = 1$. Individual orders or pairing membership accept; primitive pairing rejects. C5 reuses this witness: there are **two unique symbolic ciphertext witness families**, not three.

C6 - post-output observation control. Without semantic evaluator correctness, choose B of order $D/2$ and $A_j = -s_j B$, with $s_0 = 0$, $s_1 = 2^{k-1}$, $a \geq 4$. Then H has order 2^{k-1} and both projected G values are O. Weakened post membership and final equality accept $t=0$ for both; it is wrong for message 2^{k-1} . This is projected-output ambiguity, not an honest-map or ciphertext collision. Exact B/H orders reject.

C7/C8/C10 - encoding/receipt controls. Relative role swaps, retagging and changed receipts/target tags fail the exact-byte premise. Freshly binding a swapped key changes the key. Detectable malformed inputs reject; universal wrong-key authentication is not promised.

C9 - normalized observation control. The x-only argument on page1 yields equality of normalized maps. The f/f normalized cross-message witness is not retained.

C11/C12 - method ceiling. Anyone supplied the correct pair $f(U)$, $f(V)$ can use the public conditional decoder; this constructs no Eval2. An EndRing/evaluator solver followed by decryption is an attack upper bound. Security needs the opposite implication via a scheme-game attacker to an EndRing solver, with a complete reduction.

Inspected evidence and open obligations

Exactly four theorem families were examined. Source evidence binds retrieval times, hashes, locators, source types, inspected scope, hypotheses and applications; the linked notes preserve the inspected statements and derivations.

1. **Pairing - primary statement/proof inspected.** Victor S.Miller, *The Weil Pairing, and Its Efficient Calculation*, J.Cryptology17(2004), Definition1, journalpp.236-237 (PDF2-3), and Proposition7/proof, pp.243-245 (PDF9-11); nondegeneracy proof pp.244-245. This primary research paper uses perfect K, coprime n,p and full geometric torsion. The determinant-to-basis implication remains internally derived. Enge2014 remains supplemental exposition; earliest Weil1940 and all proof dependencies are not claimed inspected. Two local printed editorial defects are recorded in the pairing note; publisher paper.
2. **Degree norm - primary statement inspected.** Helmut Hasse, *Zur Theorie...III*, J.reine angew.Math.175(1936), §1 eq.(1), p.193; initial proof pp.193-195. Milne's second-edition monograph II§6 supplements dual/multiplicativity and the parallelogram lemma; it remains typed as a monograph. Hom transfer and the 4D bound are derived, not misquoted. Norm note.
3. **Normalized quotient - formulas inspected; general uniqueness partial.** Vélú, *Isogénies entre courbes elliptiques*, C.R.Acad.Sc.Paris273(26July1971), pp.238-241: original French scan images, §3 eqs.(8)-(13), Remark2 differential equality and Remark3 descent. Specialize the formulas to a rational order-two point; composing a steps gives normalized separable degree D. Sutherland2025 Theorems5.11/5.13 provide explicit supplemental exposition, but the original full categorical existence/uniqueness binding remains missing. Fixed-short-model rigidity is derived separately. Quotient note.
4. **Counting - primary statement inspected.** Schoof1985, Math.Comp.44(170), §1 p.483, §2 setup p.484, §3 p.490: deterministic cardinality for characteristic other than 2,3, with supplied field presentation. It explicitly does not compute full group structure. The prime-power point-order tests are separate elementary derivations; counting does not certify a basis, scalar Frobenius or End(C). Counting note; original author-hosted scan.

[Limit] Reading source statements/relevant proofs is not independent review or formal verification. Strict source gaps above remain partial; no future J5 outcome is asserted.

[O1-O6 all open] O1 still lacks the exact hard-problem sampler/distribution and complete evaluable End(E) generation. O2 lacks BaseGen, recipient map and Eval2. O3 lacks full correctness/failure bounds, especially epsilon_eval. O4 lacks the efficient private lift and costs. O5 lacks the correctly directed game reduction. O6 lacks the complete adversarial view, repeated-use/side-channel/game analysis and authentication. See the exact obligation deltas.

[Accounting] Scientific0; implementation0; formalizer0; experiment0. No Eval2 search or speed claim. dominated_by: n/a (no result claimed). sota_delta: no attack; conceptual/source/control correction only. No closures, security, novelty, practicality, deployment or finalist claim.

One routing specification, current rendered view

The routing specification fixes every node identity, label, edge direction and control disposition. The editable Mermaid is generated from it; the official Mermaid library renders the SVG. The current SVG is converted to vector PDF and included here, without a hand-drawn replacement. The graph shows evidence/control routing; its arrows are not isogenies or security reductions.

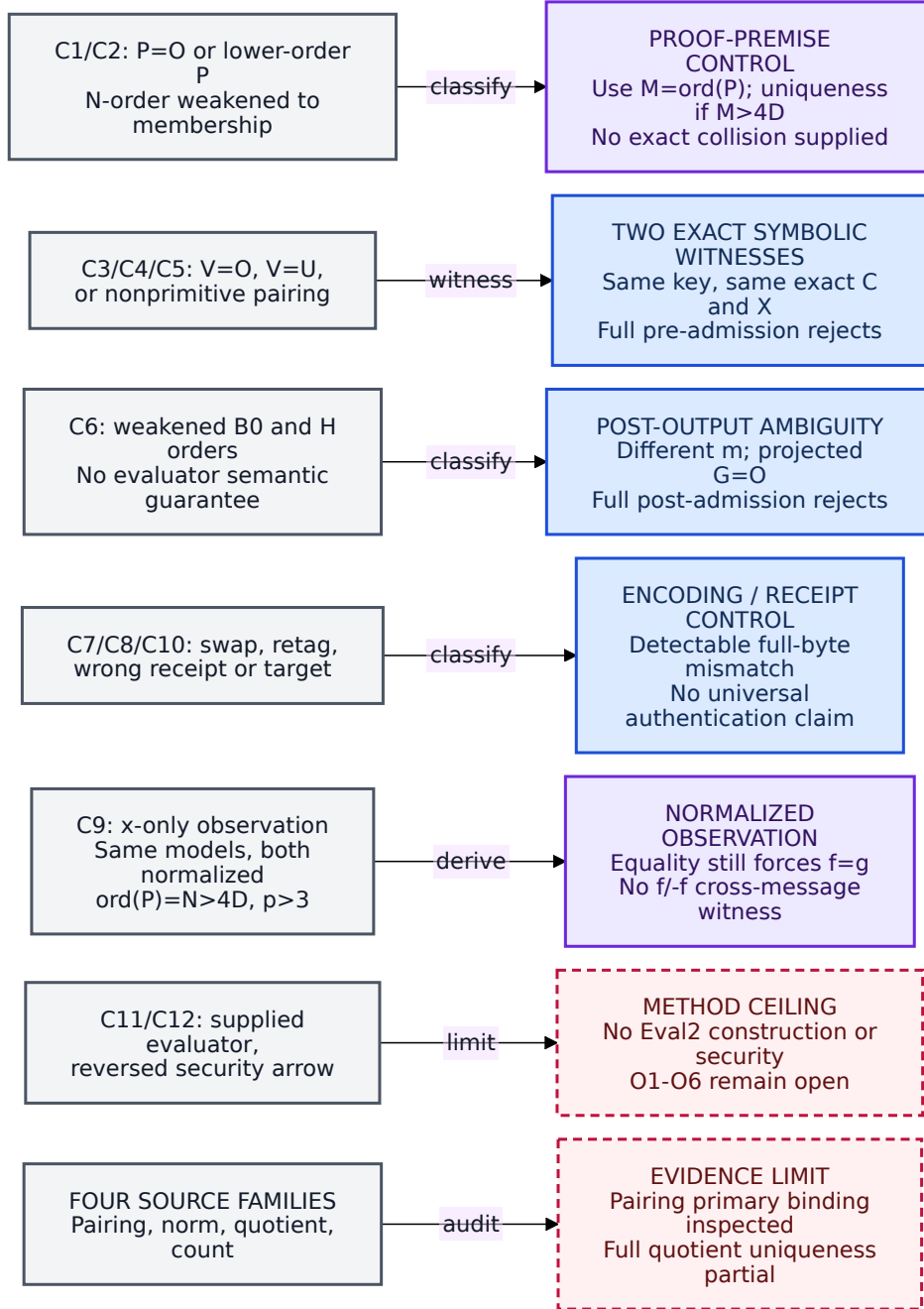


Figure 1: Control classifications and source limits

No quantitative graph applies: zero data, samples, measured costs, run IDs or uncertainty intervals. Historical graph bytes remain unchanged; this new scoped view corrects their control semantics. Actual SVG and every PDF page are visually inspected and their exact hashes bound in provenance. A fresh reviewer verdict remains separate from producer completion.